

NB37: MASTER Panel - Kalibre Stacking Raporu

SEED=42 | TEST_SIZE=0.2 | 4 kol x 5 cesitli base + Isotonic

0. KIYAS: NB14 (%50/50 yanilsamasi) vs NB36 vs NB37

NB14'un %50/50 F1~0.89'u TRAIN-DENGESI YANILSAMASIDIR: egitim ~%73 pathogenic; final test ise ~%80 BENIGN. Train dengesinde olculen F1 final dagilimi YANSITMAZ. Gercek metrik = %80/20 bootstrap pathogenic-F1.

Bu notebook (NB37) ayni kalibre+cesitli stacking'i HEM %50/50 HEM %80/20 raporlar.

Notebook	Yontem	%50/50 F1	%80/20 F1 (gercek)
NB14	OOF stack (7 base), %50/50 only	~0.89	OLCULMEDI (eksik)
NB36	Baseline (no calib stack)	~0.60	~0.60 (rapor edildi)
NB37	Kalibre+cesitli stack (no_fe_dahil)	0.8321	0.5776 [0.507-0.647]

1. 4-KOL KARSILASTIRMA (no_fe/with_fe x dahil/robust)

Birincil metrik = %80/20 bootstrap pathogenic-F1. FE ve robust kollari GERCEKTEN egitilir.

Kol	n_feat	F1_8020	CI_95	MCC_8020	Prec_8020	Rec_8020	F1_5050	gap	missFI%
no_fe_dahil	426	0.5776	[0.51,0.65]	0.465	0.462	0.775	0.832	+0.003	0.9
with_fe_dahil	440	0.5853	[0.51,0.67]	0.473	0.485	0.744	0.818	+0.000	0.9
no_fe_robust	277	0.5890	[0.50,0.65]	0.477	0.509	0.705	0.796	-0.019	0.0
with_fe_robust	291	0.5782	[0.51,0.67]	0.463	0.500	0.690	0.790	-0.007	0.0

EN IYI KOL (%80/20 F1): no_fe_robust

FE etkisi (with_fe - no_fe, dahil): +0.0077 -> FE FAYDALI

Robust etkisi (robust - dahil, no_fe): +0.0114 (robust kayipsiz, guvenli teslim)

2. Leakage Analizi (is_missing_* flag importance)

is_missing_* flag importance (dahil kollar):

no_fe_dahil : 0.9% (top: ['is_missing_AL_16', 'is_missing_AL_1', 'is_missing_AL_183'])

with_fe_dahil: 0.9%

Robust kollarda is_missing flagleri CIKARILDI (importance=0 beklenir).

GENEL LEAKAGE KARARI: DUSUK - leakage'a yaslanma sinirli (max is_missing_FI=0.9%, alarm esigi %15).

Yarisma test eksiklik oruntu su bilinmedigi icin robust kol teslim yedegi olarak tutulmal.

3. Yontem

Base Models (5 cesitli): LightGBM, CatBoost (native cat), XGBoost, LogisticRegression(L2), kNN.

OOF: 5-fold StratifiedKFold. Isotonic kalibrasyon: her base OOF train uzerinde fit.

Meta-Learner: LogisticRegression(C=1.0, L2, class_weight=balanced) -- GBM/agac meta KULLANILMADI.

Threshold: 3 mod (f1_raw / f1_8020 / mcc_8020); birincil = f1_8020 (%80/20 havuzda secildi).

Missing: M3 (>50% NaN icin is_missing_* flag + medyan imputation, orijinal korunur).

FE (with_fe kollari): EK meta (mean/max/min/std/range/disagreement/n_pos) + Grantham/BLOSUM62 (src/features.py tablolari) --

NB36 Cell 12 ile birebir.

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4. Gorseller

